RoboCupJunior General Rules

License: This document is licensed under the Creative Commons Attribution-NonCommercial-ShareAlike 4.0 International (CC BY-NC-SA 4.0) license. RoboCupJunior Regional committees are free to share and adapt this material for non-commercial, educational purposes, provided appropriate credit is given to RoboCupJunior and any derivative works are distributed under the same license. Exceptions may be granted by RoboCupJunior on a case-by-case basis.
© RoboCupJunior — https://junior.robocup.org

These are the RoboCupJunior General Rules. They apply to every competition that uses this rulebook, including the International RoboCupJunior competition. Under the license above, the organizing committee of a SuperRegional, Regional, or local competition may adapt them as needed for its competition. Check with the organizers of each competition which rules they use.

If you are unsure about any part of the General Rules or the League Rules, you are encouraged to ask on the official RoboCupJunior Forum: https://junior.forum.robocup.org/

You can also ask questions about the rules, or about RoboCupJunior in general, on the official Discord Server (https://robocup-junior.github.io/soccer-rules/discord/).

Getting Started

If you are new to RoboCupJunior, the Getting Started guides on the Official Website (https://junior.robocup.org/gettingstarted/) explain everything from building your first robot to entering your first competition:

1. **Starting a Team** (https://junior.robocup.org/robocup-beginners-guide/): form a team, build a robot, keep costs down, and find tutorials for your league.
2. **Qualification and Competitions** (https://junior.robocup.org/qualification/): how the season works, how regions qualify teams, and how to enter competitions outside your region.
3. **Your First Competition** (https://junior.robocup.org/your-first-competition/): what to expect at a competition, what to bring, and what to do when something goes wrong.
4. **For Educators** (https://junior.robocup.org/for-educators/): why robotics works in schools, and how to start a program with your students.

Team Requirements

Team Definition

Team: A team is its robot(s) and the students who build and develop them. The team members are the students who design, build, and program the robot(s) they enter. Membership may change between competition levels as described in Team Changes.

Technical Roles: Every team member must have a defined technical role, for example, mechanical design, electronics and sensors, or software. During technical judging, each member must be able

to explain their role and what they contributed to the robot(s). Non-technical tasks, such as documentation, the poster, graphic design, or the presentation, can be shared by the team, but they do not count as a technical role.

Shared Members and Robots: Team members and robots must not be shared between teams.

Age Requirements

Junior Student Members: Must be 14 to 19 years old (inclusive) on July 1st of the competition year.

Junior Mentors and Parent/Chaperones: Must be 19 years or older on July 1st of the competition year.

Entry Leagues: Entry leagues and other "Primary" divisions are not part of the International competition, but many Regional and SuperRegional competitions run them. Their organizers set the age limits, which may differ from the range above.

Team Size

Minimum Team Size: A team must have at least 2 members.

Maximum Team Size:

- Soccer and Rescue Leagues: 4 members.
- OnStage League: 5 members.

Regional and SuperRegional competitions may set their own team sizes, for example, because of venue space. At a higher-level competition (a SuperRegional or International competition the team qualified for), a team must follow that competition's maximum team size, even if the team was larger at its qualifying competition.

Team Changes

Team membership may change between competition levels, for example, when a member can no longer take part. At a higher-level competition, at least half of the team (rounded up) must be students who were members when the team qualified for that competition. New members must take part in designing, building, or programming the robot(s) before they compete, and must meet the Technical Roles requirement. A qualification and the robot(s) belong to the team and cannot be given to another team. How a region reassigns a place that a team does not use is part of the qualification process (https://junior.robocup.org/qualification/).

Team Supervision

Junior Mentor: A Junior Mentor is a supervisor who guides the team and meets the Age Requirements. Junior Mentors are not team members. They must not design, build, or program the robot(s) for the team (see Mentoring and Onsite Assistance).

Junior Mentor Requirement: Every Junior team must have at least one Junior Mentor who is registered with the team, attends the competition with the team, and can be reached by the team

and the event organizers throughout the competition.

Junior Mentors and Parent/Chaperones are responsible for supervising their team and have a duty of care for their team members' well-being, as required by their home region's rules. Any concern about a team member's well-being should be reported to the event organizers immediately.

Unless the event organizers decide otherwise, Junior Mentors and Parent/Chaperones may not enter the competition areas during rounds and interviews. They must not pressure or intimidate teams, judges, or the judging process, and must not interfere with robots. Event organizers will handle inappropriate behavior, which may lead to disciplinary action.

Robot Requirements

Robot Communication

Permitted Communication: During competition rounds, robots may communicate with each other only on the 2.4 GHz band. Their transmission power must never exceed 100 mW EIRP (Effective Isotropic Radiated Power).

Responsibility: Teams are responsible for their robots' communication. Their wireless equipment, and how they use it, must follow the laws of the host country. Spectrum availability is not guaranteed.

Component Communication: Parts of the same robot may communicate with each other.

League Adaptability: Each league may change these communication rules to fit its needs.

Safety and Power Requirements

Electrical Power:

- Robots must not be powered from a wall socket (mains electricity).
- Maximum voltage: 48 V DC or 25 V AC RMS (Root Mean Square).
- Voltage must be easy to measure during inspections. Measuring points must be covered or designed to be safe.

Battery Safety:

- Lithium batteries must be stored in safety bags, and charging must be supervised by team members in competition areas.
- Teams must follow the safety procedures, including how to handle a battery fire and how to evacuate.

Robot Safety Design:

- **Power Management**: Batteries must be firmly mounted and wiring must be safe. Every robot must have an easy-to-reach way to cut its power immediately in an emergency.
- **Mechanical Safety**: Robots must not have sharp edges, places where fingers can get caught, or

other hazards. Actuators must suit the robot's size and purpose.

- **Hazardous Behavior**: If a robot could behave dangerously, the team must tell the event organizers at least two weeks before the competition. If the team discovers such a behavior later, it must tell the organizers immediately and must not use that behavior in competition until the organizers approve it.

Documentation and Sharing Requirements

RoboCupJunior Team Posters

Purpose: Posters help teams share their robot designs and ideas with judges, other teams, and the public. Posters are displayed in the venue's public competition areas, and RoboCupJunior shares digital copies or photos after the competition.

Size: Posters must be no larger than A1 (59.4 x 84.1 cm).

Content: Posters should summarize the team's design documents and present the robot's capabilities attractively.

Technical Description Video (See League Documentation)

Content:

- **Robot Demonstration**: Show the fully working robot systems and highlight their technical features.
- **Design Process**: Explain the team's design choices and how the team solved problems.
- **Presentation**: The video should be clear and of good quality, and should explain any innovative or unusual techniques.
- **Innovation & Sustainability**: Show new technologies and sustainable practices.

Submission: Each league's guidelines will specify the video length and the deadlines.

Bill of Materials (BOM)

Submission: Teams must submit a Bill of Materials (BOM) that lists the main components and materials of their robot(s).

Details: For each component, the BOM must include:

- Name or description (for example, the part number).
- Supplier or source (including PCBs and machined parts).
- Whether it is new or reused.
- Whether it comes from a kit or is custom-built.
- Price.

Template: For the International competition, a standard BOM template will be provided with the

league documentation submissions.

Sharing Team Resources

Sharing: Materials that teams submit as part of their documentation will be shared in the leagues' GitHub repositories: https://github.com/robocup-junior

Credit: Teams must give credit to the creators of any outside work they use and follow its license. The focus should stay on the team's own learning.

Plagiarism Guidelines

External Code Use: Teams may use code written by others, but must give credit to the original authors.

Learning Priority: Teams should focus on learning and should not use complete solutions made by others. Always check the license.

Acceptable Uses of AI

Teams must disclose any generative AI tools they used to create content in their code or documentation (for example, text, figures, or images). Teams must name each AI system they used and briefly explain how they used it. Using AI for editing, grammar correction, or translation into English is common practice and does not need to be disclosed.

Spirit and Behavior

Behavior

All participants are expected to be respectful and polite to everyone, including other participants, volunteers, referees, the organizers of all Junior and Major Leagues, and the host venue.

Code of Conduct

All organizers, volunteers, team members, mentors, supporters, and visitors must follow the RoboCup Code of Ethics and Conduct (https://robocup.org/conduct). Anything that breaks the Code must be reported in the way the Code describes. The Code gives the formal reporting address (conduct@robocup.org) and explains what to do at the competition, including in an emergency.

Mentoring and Onsite Assistance

Help from other teams, mentors, teachers, parents, sponsors, and online communities is an important part of how teams learn and grow.

For fair competition and the best learning, this help must never do the team's competition work for them. A good test: the team can explain not only what each part of their robot does, but also how it works.

Teams Onsite

- During the competition, only official team members may represent the team at registration, during setup, and in the competition areas during rounds and interviews.
- At least 2 team members must be present at the competition. If fewer can attend, the team must contact the event organizers as early as possible. The organizers decide whether and how the team may compete. Unless the organizers decide otherwise, a team that competes with fewer than 2 members present is not eligible for awards or for SuperTeam participation.
- Teams are responsible for making sure their members are at the right place at the right time for every scheduled activity.
- During the competition, teams must not receive remote help with their robot, code, or strategy from anyone outside the team, for example, by phone, video call, messaging, or remote desktop. Help on site is covered by Mentoring and Onsite Assistance.
- Teams that break these rules may face disciplinary action.
- If teams have problems on site, they are encouraged to ask other teams or the organizers for help.

Violations

The event organizers may disqualify teams, team members, Junior Mentors, or Parent/Chaperones from the competition and ask them to leave the venue. This can happen for repeated or serious unacceptable behavior, or for repeated or serious violations of the General Rules or League Rules.